

Variational Methods

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Estimate ground-state energy ϵ_0 for Qm system.

Trial wavefunction $|\psi\rangle$ with $\langle\psi||\psi\rangle$

Evaluate $E_{\text{Trial}}=\langle\psi||\psi\rangle$. E_{trial} is an upper bound on E_0 with $E_{\text{trial}} \geq E_0$.

Assume true energy eigenstates are $\{|\chi\rangle, E_k\}$. $k = 0, 1, \dots$

$$|\psi\rangle = \sum_k^\infty c_k |\chi_k\rangle \quad (1)$$

$$\langle\psi||\psi\rangle = \sum_{k,l}^\infty c_l^* c_k \langle\chi_l||\chi_k\rangle = \sum_k^\infty |c_k|^2 = 1 \quad (2)$$

$$\hat{H}|\psi\rangle = \sum_k^\infty c_k E_k |\chi_k\rangle \quad (3)$$

$$\langle\psi|\hat{H}|\psi\rangle = \sum_{k,l}^\infty c_l^* c_k E_k \langle\chi_l||\chi_k\rangle = \sum_k^\infty |c_k|^2 E_k = E_{\text{trial}} \quad (4)$$

Suppose

$$|\psi\rangle \propto |\chi_0\rangle a + \epsilon |\chi_1\rangle \quad (5)$$

Normalise

$$|\psi\rangle = \frac{1}{\sqrt{1+\epsilon^2}} |\chi_0\rangle + \frac{\epsilon}{\sqrt{1+\epsilon^2}} |\chi_1\rangle \quad (6)$$

$$|\chi_0\rangle + \epsilon |\chi_1\rangle + O(\epsilon^2) \quad (7)$$

$$\langle\psi|\hat{H}|\psi\rangle = E_0 + \epsilon^2 E_1 \quad (8)$$

Suppose we have the example $\psi_\alpha(r) \propto e^{-\frac{r}{\alpha}}$

$$E_{\text{trial}}(\alpha) = \langle\psi(\alpha)|\hat{H}|\psi(\alpha)\rangle \quad (9)$$

We want to find the global min

$$\frac{dE_{\text{trial}}(\alpha)}{d\alpha} \Big|_{\alpha^*} = 0 \quad (10)$$

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